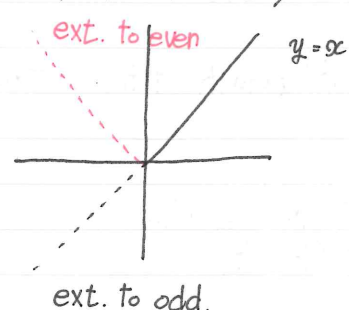


Chapter 16 Reflection methods and spherical mean.

We consider 1-dim wave equation in a half sp.

$$(16.1) \quad \begin{cases} \frac{\partial^2 u}{\partial t^2}(x, t) - \frac{\partial^2 u}{\partial x^2}(x, t) = 0, & x > 0, t \in \mathbb{R} \\ u(0, t) = 0, & t \in \mathbb{R} \\ u(x, 0) = g(x), & x > 0 \\ \frac{\partial u}{\partial t}(x, 0) = h(x), & x > 0. \end{cases}$$

<Reflection method>



Define

$$\begin{aligned} \tilde{u}(x, t) &= \begin{cases} u(x, t) & x > 0, \\ -u(-x, t) & x \leq 0 \end{cases} \\ \tilde{g}(x) &= \begin{cases} g(x) & x > 0 \\ -g(-x) & x \leq 0 \end{cases} \\ \tilde{h}(x) &= \begin{cases} h(x) & x > 0 \\ -h(-x) & x \leq 0 \end{cases} \end{aligned}$$

If u is a sol of (16.1)

$$(16.2) \quad \begin{cases} \frac{\partial^2 \tilde{u}}{\partial t^2}(x, t) - \frac{\partial^2 \tilde{u}}{\partial x^2}(x, t) = 0 & x \in \mathbb{R}, t > 0 \\ \tilde{u}(0, t) = 0 \\ \tilde{u}(x, 0) = \tilde{g}(x) & x \in \mathbb{R} \\ \frac{\partial \tilde{u}}{\partial t}(x, 0) = \tilde{h}(x) & x \in \mathbb{R} \end{cases}$$

By the d'Alembert formula.

$$\tilde{u}(x, t) = \frac{1}{2} \left(\tilde{g}(x+t) + \tilde{g}(x-t) \right) + \frac{1}{2} \int_{x-t}^{x+t} \tilde{h}(y) dy$$

Plugging the def of $\tilde{u}, \tilde{g}, \tilde{h}$

$$(16.3) \quad u(x, t) = \begin{cases} \frac{1}{2} \left(g(x+t) + g(x-t) \right) + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy & (x-t > 0) \\ \frac{1}{2} \left(g(x+t) - g(x-t) \right) + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy & (x-t < 0) \end{cases}$$

($\because \int_{x-t}^{x+t} \tilde{h}(y) dy = \int_{x-t}^{x+t} h(y) dy + \int_{x-t}^{x-t} \tilde{h}(y) dy$)

Therefore (16.3) is a sol. formula for (16.1)

<3-dim wave eq.: spherical mean>

We consider 3-dim wave eq.

$$(16.4) \quad \begin{cases} \frac{\partial^2 u}{\partial t^2}(x, t) - \Delta u(x, t) = 0, & x \in \mathbb{R}^3, t \in \mathbb{R} \\ u(x, 0) = g(x), & x \in \mathbb{R}^3 \\ \frac{\partial u}{\partial t}(x, 0) = h(x), & x \in \mathbb{R}^3 \end{cases}$$

$g = g(x), h = h(x): \mathbb{R}^3 \rightarrow \mathbb{R}$ given.

Today's aim: Derive sol. formula for (16.4)

For $x \in \mathbb{R}^3, r > 0, t \in \mathbb{R}$

define

$$U(r, t) := \frac{1}{4\pi r^2} \int_{\partial B_r(x)} u(y, t) d\sigma(y)$$

$$G(r, t) := \frac{1}{4\pi r^2} \int_{\partial B_r(x)} g(y) d\sigma(y)$$

$$H(r, t) := \frac{1}{4\pi r^2} \int_{\partial B_r(x)} h(y) d\sigma(y)$$

↑ area of $\partial B_r(x)$.

We call U, G, H spherical mean of u, g, h respectively

↑ 球面平均

Lem 16.1 (Euler - Darboux - Poisson eq.).

Let u be a sol of (16.4).

Let U, G, H are spherical mean of u, h, g respectively. Then

$$\frac{\partial^2 U}{\partial t^2}(r, t) - \frac{\partial^2 U}{\partial r^2}(r, t) - \frac{2}{r} \cdot \frac{\partial U}{\partial r}(r, t) = 0, \quad r > 0, t \in \mathbb{R}$$

$$U(r, 0) = G(r), \quad r > 0$$

$$\frac{\partial U}{\partial r}(r, 0) = H(r), \quad r > 0.$$

See Notes for precise proof.

Outline of proof: How to compute?

First from the proof of thm 10.1

$$\int_{B_r(x)} \Delta u(y, t) dy = r^2 \frac{\partial}{\partial r} \left(\frac{1}{r^2} \int_{\partial B_r(x)} u(y, t) d\sigma(y) \right)$$

$$\Rightarrow \frac{1}{4\pi r^2} \int_{B_r(x)} \Delta u(y, t) dy = \frac{\partial U}{\partial r}(r, t)$$

Idea to compute $\frac{\partial^2 U}{\partial r^2}$:

Change of variable $y = x + rz$.

$$y \in B_r(x) \Leftrightarrow z \in B_1(0)$$

$$dy = r^3 dz$$

Then

$$\begin{aligned} \frac{\partial^2 U}{\partial r^2} &= \frac{\partial}{\partial r} \left(\frac{1}{4\pi r^2} \int_{B_r(x)} \Delta u(y, t) dy \right) \\ &= \frac{\partial}{\partial r} \left(\frac{r}{4\pi} \int_{B_1(0)} \Delta u(x + rz, t) dz \right) \\ &\quad \uparrow \text{change of variable} \end{aligned}$$

Chapter 17: Kirchhoff formula.

$$\begin{aligned} &= \frac{1}{4\pi} \int_{B_1(0)} \nabla(\Delta u)(x + rz, t) \cdot z dz \\ &\quad + \frac{1}{4\pi} \int_{B_1(0)} \Delta u(x + rz, t) dz \\ &= \frac{1}{4\pi r^3} \int_{B_r(x)} \nabla(\Delta u)(y, t) \cdot \frac{y - x}{r} dy \\ &\quad \uparrow \text{change of variable} \quad + \frac{1}{4\pi r^3} \int_{B_r(x)} \Delta u(y, t) dy \end{aligned}$$

By Gauss' thm. we obtain Lem 16.1 (See Notes).

Chapter 17

From Lem 16.1

$$\frac{\partial^2 U}{\partial t^2}(r, t) - \frac{\partial^2 U}{\partial r^2}(r, t) - \frac{2}{r} \cdot \frac{\partial U}{\partial r}(r, t) = 0$$

Put $W(r, t) := rU(r, t)$

$$\Rightarrow \frac{\partial W}{\partial r}(r, t) = U(r, t) + r \cdot \frac{\partial U}{\partial r}(r, t)$$

$$\frac{\partial^2 W}{\partial r^2}(r, t) = \frac{\partial U}{\partial r}(r, t) + \frac{\partial U}{\partial r}(r, t) + r \frac{\partial^2 U}{\partial r^2}(r, t)$$

$$\Rightarrow = r \frac{\partial^2 U}{\partial r^2}(r, t) + 2 \cdot \frac{\partial U}{\partial r}(r, t)$$

$$\frac{\partial^2 W}{\partial r^2}(r, t) = r \frac{\partial^2 U}{\partial r^2}(r, t)$$

Then

$$\frac{1}{r} \cdot \frac{\partial^2 W}{\partial t^2}(r, t) - \frac{1}{r} \frac{\partial^2 W}{\partial r^2}(r, t) = 0$$

$$\Rightarrow \frac{\partial^2 W}{\partial t^2}(r, t) - \frac{\partial^2 W}{\partial r^2}(r, t) = 0, \quad r > 0$$

$$W(0, t) = 0$$

Therefore we can use the sol. formula (16.3) to W

For $0 < r \leq t$, we apply (16.3) then

$$W(r, t) = \frac{1}{2} \left(G(r+t) - G(t-r) \right) + \frac{1}{2} \int_{t-r}^{r+t} H(y) dy$$

Recall $W(r, t) = r U(r, t)$

$$= \frac{r}{4\pi r^2} \int_{\partial B_r(x)} u(x, t) d\sigma(y)$$

Hence

$$U(r, t) = \frac{G(r+t) - G(t-r)}{2r} + \frac{1}{2r} \int_{t-r}^{r+t} H(y) dy$$

Letting $r \downarrow 0$ we have, $t+r$

$$u(x, t) = \lim_{r \downarrow 0} \left(\frac{\tilde{G}(r+t) - \tilde{G}(t-r)}{2r} + \frac{1}{2r} \int_{t-r}^{r+t} \tilde{H}(y) dy \right)$$

$$= \tilde{G}'(t) + \tilde{H}(t)$$

$$= \frac{\partial}{\partial t} \left(\frac{1}{4\pi t^2} \int_{\partial B_t(x)} g(y) dy \right) \quad \text{--- } \overset{= G(t)}{\text{---}}$$

$$+ \frac{1}{4\pi t^2} \int_{\partial B_t(x)} h(y) dy$$

To compute $\frac{\partial}{\partial t} \tilde{G}(t)$. Consider the change of variable.

$$y = x + tz, \quad y \in \partial B_t(x) \leftrightarrow z \in \partial B_1(0)$$

$$d\sigma(y) = t^2 d\sigma(z)$$

$\uparrow \partial B_t(x)$ is 2-dim.

$$\frac{\partial}{\partial t} \left(\frac{1}{4\pi t^2} \int_{\partial B_t(x)} g(y) d\sigma(y) \right)$$

$$= \frac{\partial}{\partial t} \left(\frac{1}{4\pi} \int_{\partial B_1(0)} g(x+tz) d\sigma(z) \right)$$

$$= \frac{1}{4\pi} \int_{\partial B_1(0)} \nabla g(x+tz) \cdot z d\sigma(z)$$

$$= \frac{1}{4\pi t^2} \int_{\partial B_t(x)} \nabla g(y) \frac{y-x}{t} d\sigma(y)$$

useless
(wrong)

Wrong computation

Problem: Where?

Ans: Applying (16.3)

$$G \rightarrow \tilde{G} = rG, \quad H \rightarrow \tilde{H} = rH$$

$\uparrow$
Initial data should be change.

$$\frac{\partial}{\partial t} \left(\frac{t}{4\pi t^2} \int_{\partial B_t(x)} g(y) d\sigma(y) \right)$$

$$= \frac{\partial}{\partial t} \left(\frac{t}{4\pi} \int_{\partial B_1(0)} g(x+tz) d\sigma(z) \right)$$

$$= \frac{1}{4\pi} \int_{\partial B_1(0)} g(x+tz) d\sigma(z) + \frac{t}{4\pi} \int_{\partial B_1(0)} \nabla g(x+tz) \cdot z d\sigma(z)$$

$$= \frac{1}{4\pi t^2} \int_{\partial B_t(x)} g(y) d\sigma(y) + \frac{1}{4\pi t} \int_{\partial B_t(x)} \nabla g(y) \cdot \frac{y-x}{t} d\sigma(y)$$

Note that $\partial B_t(x) = \{y \in \mathbb{R}^3; |x-y| = t\}$ Hence

$$(17.3) \quad u(x, t) = \frac{1}{4\pi t^2} \int_{|x-y|=t} \left(t h(y) + g(y) + \nabla g(y) \cdot (y-x) \right) d\sigma(y)$$

Thm 17.1 (Kirchhoff's formula)

For $g \in C^3(\mathbb{R}^3)$, $h \in C^2(\mathbb{R}^3)$ let u be a func. defined by (17.3) Then u is a sol of (16.1)